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A Robust Method for Automatic and Dynamic Acquisition of Japanese Syntactic Rules from Raw Text Using Inductive Learning

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ABSTRACT
We propose a robust method for automatic and dynamic acquisition of Japanese syntactic rules from raw text using inductive learning. The previous methods have not had the robustness, that is, the capability of unsupervised learning, a dynamic adaptation to a current domain, and an acquisition from the condition of no initial rule. We describe the framework for dynamic unsupervised learning and the learning algorithm. Then, we demonstrate its robustness in the experiments.

KEY WORDS
Natural Language Processing, Inductive Learning, Unsupervised Learning, Context-Free Grammar

1. Introduction

There are many rule-based methods for syntactic analysis. The performance of such methods is limited by the quality of the used rules. It is necessary for a robust analysis system to give rules that can be applied to any domains. As it is difficult to manually give such rules, we consider that the system must automatically acquire them.

We consider that it is effective for natural language processing to imitate a language processing of human beings, because natural languages have inevitably borne by social communication among human beings. Language acquisition is an important factor of the language processing of human beings. An infant can acquire any language as his mother tongue, and learn the implicit grammar although his parents can not directly teach him it. Furthermore, he will learn new words and grammatical knowledge step by step with the years, while he uses the knowledge for understanding or dialogue. This is a very robust capability. If a system has the capability, the system will be able to acquire syntactic rules automatically and dynamically.

In this paper, we propose a method for the automatic and dynamic acquisition of Japanese syntactic rules from raw text corpora using inductive learning. This method does not depend on only Japanese, although we conducted experiments in Japanese that is the mother tongue of the authors.

Many methods for automatic acquisition of rules have been researched[1, 2, 3, 4, 5, 6, 7]. However, these methods have problems as follows from the standpoint of constructing a robust system.

The methods of Hwa[1], Shirai et al.[2], and Brill[3] are using supervised learning from bracketed corpora. Although bracketed corpora have appeared of high quality and quantity, they do not always fit the domain a user wants. It takes much labor for a user to bracket raw sentences, and not all users have knowledge about correct bracketing. Therefore, a robust system must have the capability of unsupervised learning. A system based on our method inductively learns from raw sentences using some heuristics, which are described in section 4, and frequency of segments such as characters, words or phrases.

The methods of Chen[4], and Lari and Young[5] can learn from raw corpora. A common problem in a corpus-based approach is that the acquired knowledge depends on a training corpus. Their learning processes are independent of an analysis process. Rules acquired by their methods are never updated even if they can not be applied to sentences that their system must analyze. This is a problem because it is pragmatically impossible to give rules beforehand that can be applied to all domains. We solve this problem by a dynamic adaptation to a domain of sentences our system analyzes in real time. If the system can not analyze an input sentence, new rules are acquired to complete the halfway-analyzed result. In section 3, we describe the problems and solution when our system dynamically learns.

The purpose of Kiyono and Tsujii's research[6, 7] is to adapt initial rules to a domain of a corpus. However, their method depends on an amount of given initial rules. Their experiment shows their method needs sufficient initial rules in order to acquire new rules. This is a problem because the supply of sufficient initial rules is as difficult as the supply of rules that can be applied to all domains. Our system can acquire new rules under any condition of initial rules even if there is no initial rule.

No initial rule means that our system has no information about dependencies between words, vocabularies, and a part of speech. We do not give information about vocabularies and a part of speech because of the following ambiguity. There is no space between words in Japanese. While how to divide a sentence into words usually causes the change of sense, there are some cases of division that cause little change. For example, in the case of "JY6/9k

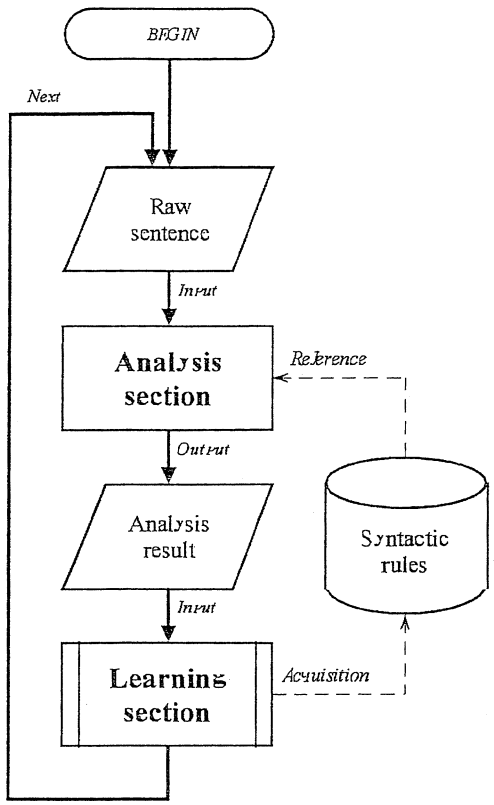


Figure 1. Outline of our method

(benkyosuru)[1]" meaning *study* in English, some Japanese regard it as a verb. Other Japanese regard it as two words that consist of a noun "JY6/(benkyo)" meaning *a study* and a verb "9k(suru)" meaning *do*. Therefore, our system acquires syntactic categories instead of symbols of a part of speech, and uses them in acquired rules. In section 5, we demonstrate the capability of acquisition from the condition of no initial rule.

The proposed method has all of the above robustness. Furthermore, our method has the following advantages. Our system can acquire rules to divide an input sentence into a word sequence. This division is a non-trivial task and can not be ignored in languages such as Japanese. Therefore, this is a great advantage. Characters we usually see do not always correspond to a sentential symbol that a parser based on top-down approach assumes as a root of a parsing tree. Since our system does not assume such a symbol, it can analyze any type of input and learn from it.

The syntactic rules in this paper are kinds of probabilistic context-free grammar rules expressed in Chomsky normal form. We choose the form because it is regarded as a standard.

[1]Characters in parenthesis represent a pronunciation of Japanese.

2. Outline

Figure 1 shows the outline of our method. The learning section is close to the analysis section, and learns from the analyzed result. Because an input sentence is shared in the learning and analysis sections, the system can dynamically adapt to a current domain.

Using acquired rules so far, the system divides a sentence to a word sequence, assigns a syntactic category to a word, and constructs a parsing tree. The parsing is basically based on a bottom-up approach because we do not assume a sentential symbol. When the system can not assign a syntactic category to a word, the system infers a syntactic category for the word by a derivation from a root of a partial parsing tree based on a top-down approach. In the case of comprehension of an ambiguous word by human beings, it has been shown all senses of an ambiguous word are briefly activated regardless of context, and then contextual information suppresses inappropriate senses [8, 9]. We suppose this same selection process is used in syntactic analysis. If several results are analyzed from a sentence, the system selects the most plausible result. The plausibility is calculated by the active value, which is described in section 3, of the rules used in the result. A segment the system can not divide is regarded as an unknown word. When the system has no syntactic rule, the system regards a whole sentence as an unknown word and outputs it. Due to space limitations, we will not specify the more detailed analysis algorithm in this page[2].

When the system can not completely analyze a sentence, the result is incomplete, that is, it includes unknown words or consists of several partial parsing trees. According to the framework described in section 3 and the algorithm described in section 4, the system acquires the rules to recognize the unknown words or to express the result by one parsing tree.

3. Framework for dynamic unsupervised learning

Because of unsupervised learning, the system must infer plausible rules from all candidates of rules the system can make. Our method uses frequency as a measure of plausibility like other statistics-based methods. A statistical approach assumes a sample is representative of the population. Because the system does not have a sufficient sample in the earlier period, the simple probability is not trusted. Therefore, we do not use a simple probability. Acquired rules are used in the analysis section without checking them by answers. If erroneous rules are applied to a sentence, the result is also erroneous. Therefore, the system must suppress the acquisition of erroneous rules as much as possible. When we use a simple frequency, a frequency of erroneous rules will be greater than the threshold after a

[2]We described the analysis algorithm in a previous paper.[10]

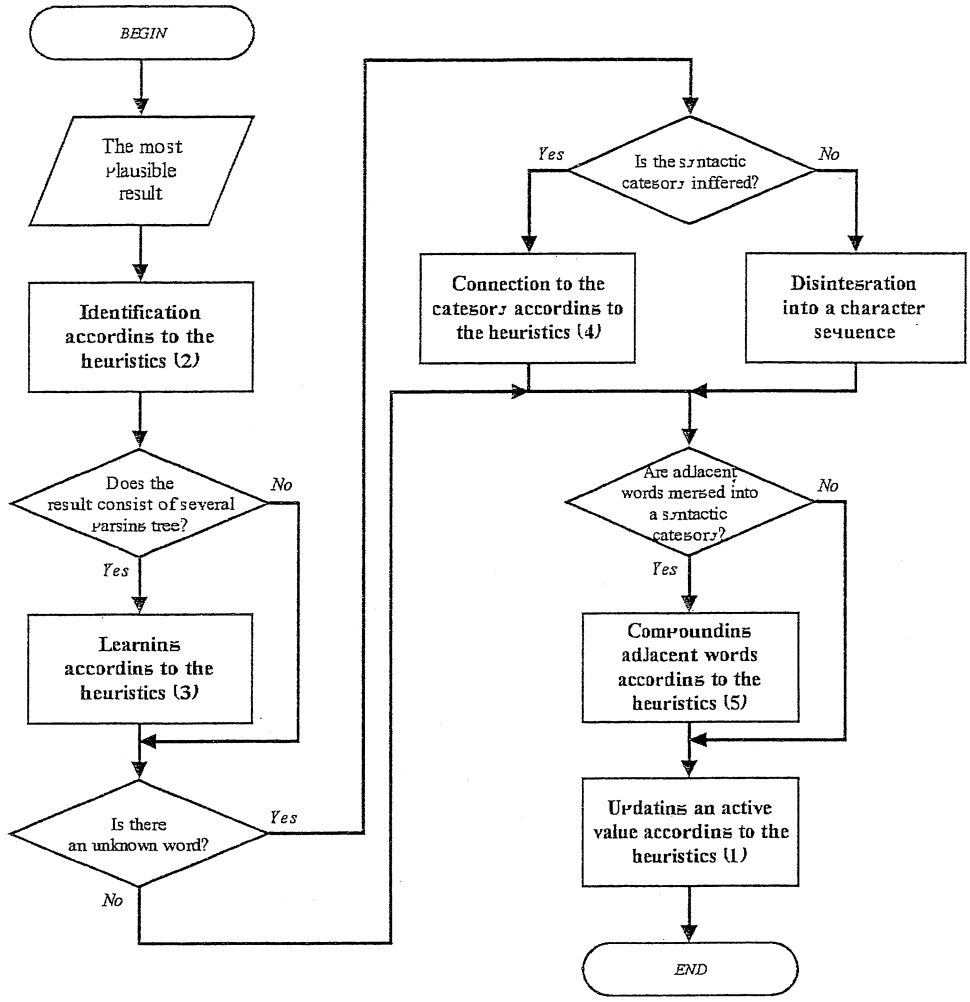


Figure 2. Processing of the learning section

tremendous number of sentences are inputted. Therefore, we do not use a simple frequency.

To solve these problems, we introduce "probation" to acquired rules. If there are several rules the system can acquire, all of them are acquired as a rule on probation. A rule on probation is different from a general rule in that it can not be applied in analysis. The system then dynamically determines general rules from the rules on probation as follows. Each rule has "an active value" instead of a frequency. A rule on probation becomes a general rule if the active value is over a certain threshold, and vice versa. When a new sentence is input, the system decreases the active value of all rules by one. A rule is removed if the active value is zero. If the system attempts to acquire a presently existing rule, or if a rule is applied in analysis, the active value of the rule is increased by a fixed value. In other words, the system selects only rules that frequently appear in a recent period. The fixed value and threshold are respectively set to 800 and 4,800 from the result of preliminary experiments.

The plausibility in analysis part is calculated by the active value, and it is different from a general probability. This is more preferred from the standpoint of adaptation to a current domain.

4. Algorithm

Figure 2 shows the processing of our inductive learning.

We use the following five heuristics for the inductive learning.

(1) Correct rules are applied to actual sentences more than erroneous rules.

(2) Two categories are the same if they are in the same surroundings.

(3) Adjacent syntactic categories can be merged into a syntactic category.

(4) A word is connected to a syntactic category if they are corresponded in a parsing tree.

(5) Adjacent words can be compounded into a word.

The framework described in section 3 is based on the heuristics (1). These heuristics are based on analogy and ststistics. It has been generally accepted that analogy is an important capability in learning of human beings. Saffran et al.[11] insisted that 8-month-old infant can compute the probability of adjacent phonemes. We consider that the capability of computation is working not only for phonemes but also for other linguistic factors. This is why we assume these heuristics.

In our system, new syntactic categories are optionally created when the system needs new categories to acquire rules. However, the increase in categories causes an increase of rules and deterioration in the quality of analysis. Therefore, the system identifies created categories with existing categories according to the heuristics (2). The surroundings are the forward and backward category sequence of the category the system attempts to identify. Ideally, it is to be desired that the category sequence corresponding to the whole sentence should be regarded as the surroundings. However, this also causes the problem of data sparseness. Therefore, the system performs the identification using the bottom category sequence of the partial parsing trees of which the root category is the same category. In this paper, we do not treat the problem of which category the acquired and identified categories correspond to (i.e. noun, verb, etc.), but will do so in a future paper.

If the result consists of several parsing trees, the system acquires rules to merge the root categories of the trees according to the heuristics (3). A rule for merging categories is referred to as a merge rule. Because we assume rules expressed in Chomsky normal form, the right-hand side of a merge rule is limited to two categories. If a result consists of more than two trees, the system can not determine which of the adjacent root categories the root category of middle trees should be merged with. According to the framework described in section 3, the system acquires all rules on probation from all pairs of the root categories. Then, general merge rules are dynamically selected.

If there is an unknown word in a result, processing is done as follows. If the syntactic category corresponding to the word is inferred from the parsing tree, the system acquires a new rule to connect the word to the category according to the heuristics (4). We refer to a rule connecting a word to a category as a word rule. Otherwise, the system disintegrates the word into a character sequence. The system acquires word rules by regarding each character as a word, and then also acquires merge rules between adjacent characters. If the sequence consists of more than two characters, the system processes in the same framework.

If two adjacent words are merged into a syntactic category, the words may be expressed by a compound word, and the system acquires a word rule on probation for the compound word according to the heuristics (5). Acquired word rules are applied for division of a sentence and for assignment of a syntactic category.

5. Result

We conducted two experiments to demonstrate robustness, that is, the capability of dynamic acquisition and adaptation under any conditions.

As measures of evaluation, the acceptance rate (AR) and correctness rate (CR) were used. AR is the percentage of acceptable results expressed by one parsing tree corresponding to a whole input sentence. If there are several results in an analysis, we evaluate the one result that has the highest plausibility, calculated by the active value of the used rules. The AR indicates to what extent the system has acquired rules that can be applied to sentences. The CR is the percentage of correct results that are the correct answer to an input sentence. The AR and CR are respectively calculated by the following expressions:

$$AR = \frac{A\ total\ of\ acceptable\ results}{A\ total\ of\ input\ sentences} \times 100\ [\%]$$
$$CR = \frac{A\ total\ correct\ results}{A\ total\ of\ input\ sentences} \times 100\ [\%]$$

The sentences used in our experiments are from a Japanese textbook for foreigners. The average length of these sentences is 12.6 characters. While native Japanese speakers can understand the sentences, they are correctly parsed a little by KNP that is one of the most popular parsers for Japanese[12]. This is because they are not written in a formal style the KNP assumes.

In the first experiment, the system learns from the condition of no initial rule. This is the most disadvantageous condition and a simulation of the system in the earlier period. We used 860 sentences in the experiment. We repeatedly input the sentences in order to observe how the system adapts itself to the domain of input sentences by dynamic acquisition. Figure 3 shows the change in the AR. We observed that the AR increases with the number of times sentences were input, and by the 30th input time the system achieved an AR of 85% with a CR of 37%. This result demonstrates the ability of dynamic acquisition from the condition of no initial rules.

In the second experiment, the system learns from the condition of being given basic initial rules. Practically, the system already learns some rules and must analyze unknown sentences excluded from past input sentences. The second experiment aimed to find to what extent the system could acquire rules under practical conditions. In the experiment, we selected 700 learning sentences and 50 test sentences from the textbook at random. The learning and test sentences did not overlap.

We prepared five sets of initial rules as follows. The first set (FS) consists of rules that can correctly analyze the learning and test sentences if rules are manually applied. The second set (SS) consisted of rules that are three quarters of rules in the FS selected at random. The third set (TS) consisted of rules that are two thirds of rules in the SS selected at random. The number of rules in the TS was half the number of rules in the FS. The system respectively trained the rules of the SS and TS using learning sentences until an AR of 80% was achieved. The rule sets at that time

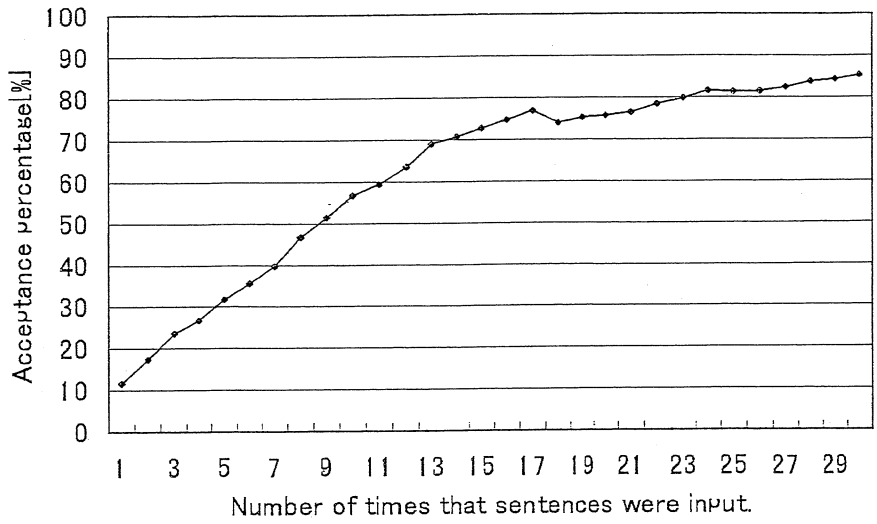


Figure 3. Change in acceptance rate of first Experiment

Table 1. Result from second Experiment

	FS	SS	SS'	TS	TS'
Number of correct results	39	29	31	13	21
Number of erroneous results	11	16	15	15	24
Number of incomplete results	0	5	4	22	5

are labeled SS' and TS' respectively. Table 1 and 2 show the obtained results when the system respectively analyzed the test sentences using the FS, SS, SS', TS and TS'. The improvement rate (IR) is the rate of improved results, and used as a measure of improvement. When we regarded the result of the FS as the ideal maximum values, the SS was improved by 20% in both AR and CR. The TS was then improved by 77% in AR and 31% in CR respectively.

6. Discussion

We can not directly compare our method to other methods in the same condition because previous methods do not have the robustness our method has. However, we describe the results of Shirai et al.[2] for a reference. Their system automatically extracted Japanese probabilistic context free grammar from 180,000 bracketed sentences in EDR corpus[13]. When the system analyzed 20,000 test sentences with the extracted grammar, a 92% acceptance rate was achieved. For the most probable parse tree, the system achieved a 54.30% brackets recall, 65.74% brackets precision and 8.47% sentence accuracy. The acceptance rate and the sentence accuracy is regarded as AR and CR respectively. While the AR our system achieved is not as great as theirs, the CR is far greater than theirs. In spite of unsupervised learning, this result shows our method is effective.

We discuss the incline of learning. In table 1, our system that learned from the TS acquired more correct rules than the system from the SS, while the TS is more incomplete than the SS. This is in contrast to the result of Kiyono and Tsujii[6]. We consider that our method is more effective because the learning process aims to complete knowledge that is lacking. This is because our system does not depend on existing rules and acquires rules only from analyzed results.

Table 2 shows that the AR grows much more quickly than the CR. Our acquisition algorithm does not cooperate with existing rules, and therefore, acquired rules tend to be correctly applied to only a specific sentence. Such rules cause an erroneous result when they are applied to other sentences. This is why the AR grows much more quickly. Table 3 shows the numbers of rules in each set. Both the SS' and TS' are larger than the FS, and the TS' is the largest. This means acquired rules can not be generalized very much. If the identification of syntactic categories works better, this problem may be solved.

While we demonstrated our method has robustness, there is room for improvement in the precision. The framework described in section 3 is the base of robustness, and does not depend on acquisition algorithms very much. Therefore, we will be able to increase precision by the im-

Table 2. Acceptance rate and correctness rate from second Experiment

	FS	SS	SS'	TS	TS'
Acceptance rate	100%	90%	92%	56%	90%
Correctness rate	78%	58%	62%	26%	42%

Table 3. The number of rules in each set

	FS	SS	SS'	TS	TS'
Number of rules	350	263	410	175	493

provement of acquisition algorithms, while the robustness is kept.

7. Conclusion

We have proposed a robust method for the automatic and dynamic acquisition of Japanese syntactic rules that fit the current domain. Because our system inductively learns new rules only from the raw text corpora using some heuristics and frequency, users' labor is greatly reduced.

We demonstrated the capability of acquisition from the condition of no initial rule in the experiment. As a result, the system achieved an AR of 85% and a CR of 37%. We also observed to what extent the system could acquire rules under practical conditions. The system analyzed unknown sentences using rules that were acquired from sentences belonging to the same domain from the condition of incomplete initial rules. As a result, the system acquired more correct rules in inverse relation to the incompleteness of initial rules.

Our proposed method, especially the framework for dynamic unsupervised learning, brought robustness the previous methods had not had. The framework may be applied to other learning system for semantic or contextual knowledge. Furthermore, our method will be applied not only to Japanese but also to other languages because the method does not depend on a specific language.

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